

Previous years' questions:

1. Draw the circuit diagram of a DRAM cell with proper labels and explain its working. [10]
2. Draw the memory hierarchy pyramid. What is the relation between speed, size and cost? [5]
3. Explain the concept of memory hierarchy with design and characteristics. [5]
4. Explain the Von Neumann architecture with a suitable diagram. [10]
5. In brief, explain the units of a computer system. [5]
6. Explain Basic Instruction Cycle with state diagram. [10]
7. What is RAM memory? Explain the working of a DRAM cell with a neat diagram. Give any four differences between SRAM and DRAM. [10]
8. Explain the bus interconnection scheme of the Von Neumann architecture. [10]
9. What are the actions taken by the CPU when instructions are fetched from memory? [5]
10. Explain the concept of interrupt handling. [10]
11. Explain the working of DRAM architecture. [10]
12. In brief, explain the functional units of a computer system.
13. Explain the basic instruction cycle with a state diagram.

Syllabus:

4.1 Von Neumann Architecture, Functional Units of Computer, Basic Instruction Cycle, Interrupts
4.2 Memory sub-system - Typical architecture of RAM and ROM, SRAM and DRAM - architecture and comparison, memory hierarchy

Von Neumann Architecture

Consists of three main components: a) CPU b) Memory Unit c) Input/output devices

Central Processing Unit

- The Control Unit (CU), the Arithmetic and Logical Unit (ALU) and a variety of registers make up the CPU.
- This is the most important part of any computer system, akin to the brain.
- It includes all the circuitry required to store the input, process the data and generate the output.
- The CPU always follows computer program instructions that tell it about what data to process and how to process the data.
- We cannot run applications on a computer without the CPU.

CU: It is responsible for processor control signals. It controls the input/output flow, how data moves throughout the system, and also gets code for instructions.

ALU: It is the unit that handles all of the CPU's computations, such as addition, subtraction and comparisons. Logical, arithmetic, and bit shifting operations are also handled by the ALU.

Registers: Registers refer to high-speed storage areas in the CPU. The data processed by the CPU are fetched from the registers.

Registers	Description
MAR (Memory Address Register)	This register holds the memory location of the data that needs to be accessed.
MDR (Memory Data Register)	This register holds the data that is being transferred to or from memory.
AC (Accumulator)	This register holds the intermediate arithmetic and logic results.
PC (Program Counter)	This register contains the address of the next instruction to be executed.
CIR (Current Instruction Register)	This register contains the current instruction during processing.
IBR (Instruction Buffer Register)	This register contains instructions that aren't to be executed immediately.

Main memory unit (a.k.a. registers)

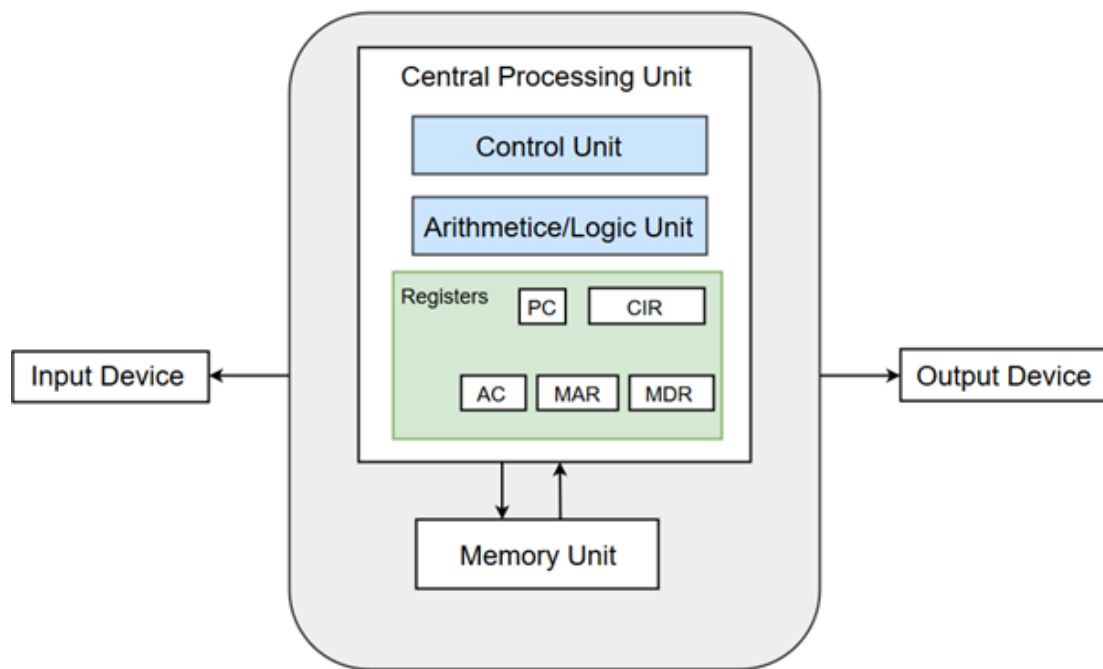
- Memory units are typically used to store the information of the programs.
- The computer has two types of storage systems: volatile and non-volatile. Volatile storage systems lose data when powered off, and non-volatile systems do not lose data when turned off.
- Memory units store non-volatile data, and they are used to store your preferences, internet bookmarks and many other items. When you turn on the computer, you can access the data stored in this memory device.

- The storage capacity of the memory unit differs depending upon the manufacturer and design.
- The memory units are used in computers because they store the frequently accessed data.

Input/output devices

Program or data is read into main memory or secondary storage from the input device under the control of the CPU input instructions. Output devices are used to present information by a computer. If some results are evaluated by the computer and are stored in the computer, then with the help of output devices, we can present them to the user.

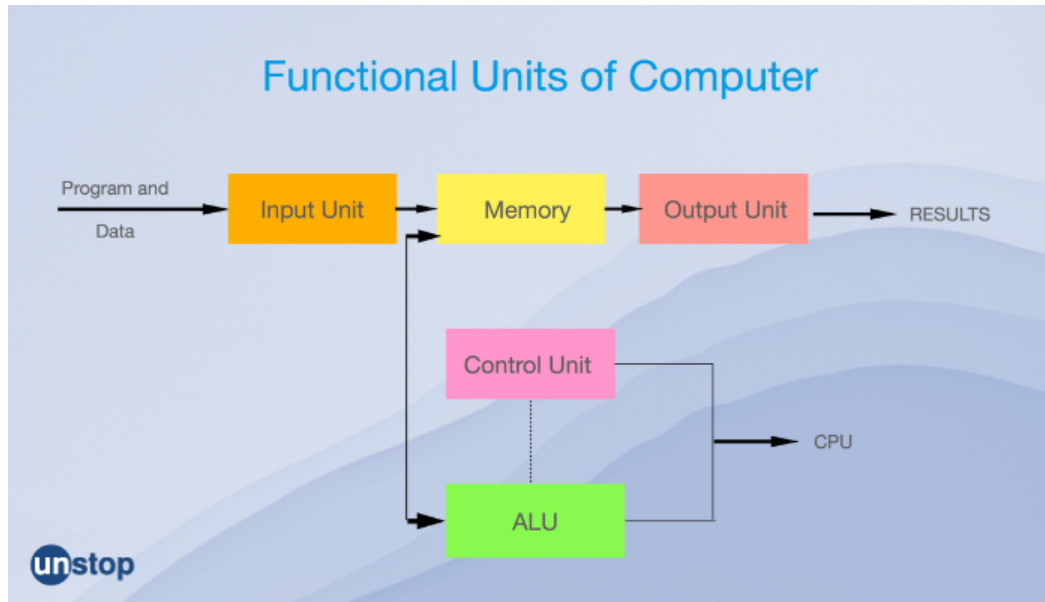
Von-Neumann Basic Structure:



Von Neumann Bottleneck

Whatever we do to enhance the performance of the computer, we can't ignore the fact that instructions can only be performed one at a time and can only be carried out sequentially. Both these factors increase the incompetence of the CPU; this condition is called the Von Neumann Bottleneck. We can provide a Von Neumann CPU with more cache and more RAM but if the original CPU is to be made better then we need to perform an inspection of the CPU configuration.

Functional Units of a Computer



Input unit

The input unit is responsible for reading data. When users give a command or when data is fed into the computer, the input unit comes into play. The input unit allows the system to understand what kind of operations are to be performed. Along with this, an input unit also determines the movement of data between the I/O devices and the computer.

The input unit uses various peripheral devices to complete its tasks. Some of the input devices include a keyboard, mouse, touchpad, trackball, etc. These devices are also known as graphic input devices as the display screen shows the actions undertaken using these input devices. For example, you can see the pointer moving on the screen when you move the mouse.

Arithmetic and Logical Unit (ALU)

The **ALU** is used for performing arithmetic operations. These arithmetic operations include addition, subtraction, multiplication, division, and comparing numbers. Additionally, it also performs logical operations, such as 'AND', 'OR', and 'NOT'.

The ALU functions by presenting the instruction to the system and storing the result in the memory. Along with this, ALU also stores the results of the operations in the registers.

Another feature of the ALU is that it has various operands. The operands are placed into the registers. The function of these operands is to store one word at a time. They help in faster retrieval of data.

Output unit

The purpose of the output unit is to process and display results. Here, the data in the binary code is converted into human language. Output devices are monitors, printers, plotters, etc.

Control unit

The control unit is an essential part of the computer as it controls data flow which is sent and received by the CPU. The control unit acts as a coordinator. In this way, the control unit is responsible for accepting the input of the data, placing it in memory, processing the stored input, and generating the output. In this way, it can be seen that the control unit is also an interpreter which decodes the information, converts it into executable data, and instructs the other components of the computer system to undertake the required action.

Memory unit

The memory unit is another component of a computer system. The main function of memory is to store data and instructions. The memory unit stores two types of data. One, it stores data associated with operations to be performed. Two, it also stores data that are related to the program. It keeps track of the interim results of the processing of data.

Memory in the memory unit is of three types: a) primary memory b) secondary memory c) cache memory.

Primary memory

Primary memory is popularly known as Random Access Memory (RAM), the main memory, or primary storage. In primary memory, one word can be stored or retrieved in one basic operation. A word can be between 16-64 bits.

It is important for the execution of programs that are placed in it. It is from here that instructions of the program are retrieved one by one by the processor for execution. Some of the features of primary memory are:

- It is the fastest accessible memory.
- It is expensive.
- It is also called "volatile memory".
- It stores a large number of semiconductor cells.

Secondary memory

The hard disk is the secondary memory of your computer. They are detachable. Hence, even if

the power supply to the computer is turned off, you will not lose data stored in secondary memory.

SSD cards, pen drives, and flash drives are some modern examples of secondary memory. Other examples of memory devices in this category include optical and magnetic disks. Some of the features of secondary memory are:

- It is slower.
- It stores a large volume of data.
- It is less expensive.

Cache memory

Also known as supplementary memory, cache memory is used for retrieving data faster than primary memory. It is used for storing data that is required frequently by the processor. In this way, it reduces the average time to access data from the main memory. Some of the features of cache memory are:

- It stores data temporarily.
- It is smaller in size than the primary memory, but is faster.

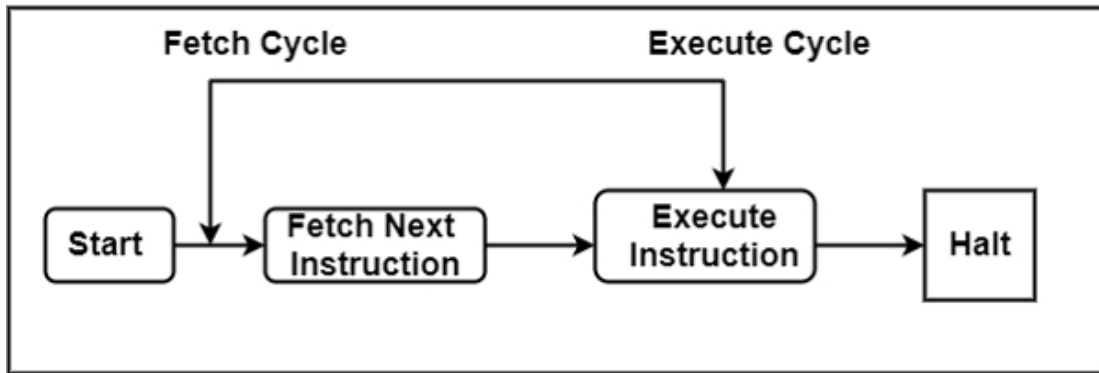
Instruction Cycle

A program residing in the memory unit of a computer consists of a sequence of instructions. These instructions are executed by the processor by going through a cycle for each instruction. In a basic computer, each instruction cycle consists of the following phases:

1. Fetch instruction from memory.
2. Decode the instruction.
3. Read the effective address from memory.
4. Execute the instruction.

After these four procedures are done, the control switches back to the first step and repeats the similar process for the next instruction. Therefore, the cycle continues until a halt condition is met. The figure shows the phases contained in the instruction cycle.

Instruction Cycle



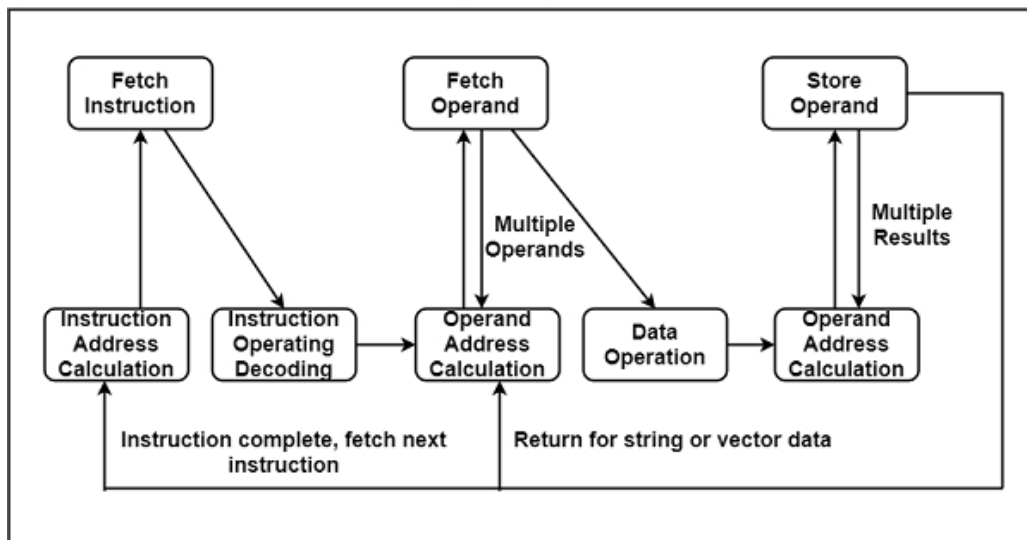
The halt condition appears when the device is turned off, or due to some unrecoverable errors.

Fetch cycle: The program counter holds the address of the instruction to be implemented. The processor fetches the indicated instruction from the memory. The current instruction is executed and the next instruction is loaded onto the PC to be read and executed.

Execute cycle: In the execute cycle, the processor implements the required operations on the data. The data transfer for implementation takes place in two methods as follows:

- 1) Processor-memory: The data sent from the processor to the memory or from the memory to the processor
- 2) Processor-I/O: The data can be sent to or from a peripheral device between the processor and the I/O device.

Instruction Cycle State Diagram

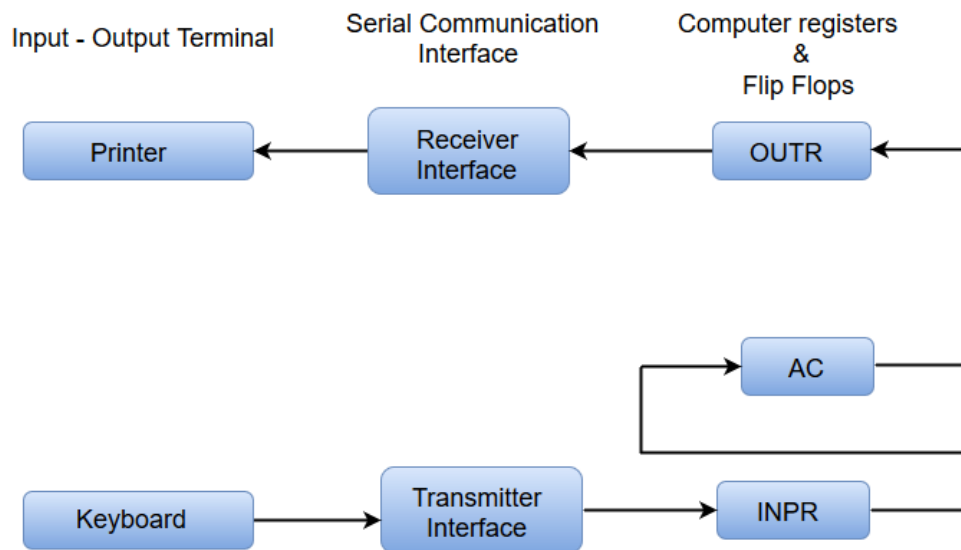


Input/Output Configuration

In computer architecture, input-output devices act as an interface between the machine and the user.

Instructions and data stored in the memory must come from some input device. The results are displayed to the user through some output device.

Input - Output Configuration:



- The input-output terminals send and receive information.
- The amount of information transferred will always have eight bits of an alphanumeric code.
- The information generated through the keyboard is shifted into an input register 'INPR'.
- The information for the printer is stored in the output register 'OUTR'.
- Registers INPR and OUTR communicate with a communication interface serially and with the AC in parallel.
- The transmitter interface receives information from the keyboard and transmits it to INPR.
- The receiver interface receives information from OUTR and sends it to the printer serially.

Interrupts

The interrupt is a signal emitted by hardware or software when a process or an event needs immediate attention. It alerts the processor to a high-priority process requiring interruption of the current working process. In I/O devices one of the bus control lines is dedicated for this purpose and is called the *Interrupt Service Routine (ISR)*.

When a device raises an interrupt at let's say process i , the processor first completes the execution of instruction i . Then it loads the Program Counter (PC) with the address of the first instruction of the ISR. Before loading the Program Counter with the address, the address of the interrupted instruction is moved to a temporary location. Therefore, after handling the interrupt the processor can continue with process $i+1$.

While the processor is handling the interrupts, it must inform the device that its request has been recognized so that it stops sending the interrupt request signal. Also, saving the registers so that the interrupted process can be restored in the future, increases the delay between the time an interrupt is received and the start of the execution of the ISR. This is called Interrupt Latency.

Software Interrupts

- Produced by a software or a system as opposed to hardware.
- Traps and exceptions are other names for software interruptions.
- They serve as a signal for the operating system or a system service to respond to an error condition.

A particular instruction known as a “interrupt instruction” is used to create software interrupts. When the interrupt instruction is used, the processor stops what it is doing and switches over to a particular interrupt handler code. The interrupt handler routine completes the required work or handles any errors before handing back control to the interrupted application.

Hardware Interrupts

In a hardware interrupt, all the devices are connected to the Interrupt Request (INTR) Line. A single request line is used for all ‘ n ’ devices. To request an interrupt, a device closes its associated switch. The value of INTR is the logical OR of the requests from individual devices.

The sequence of events involved in handling an IRQ:

- Devices raise an IRQ.
- The processor interrupts the program currently being executed.
- The device is informed that its request has been recognized and the device deactivates the request signal.
- The requested action is performed.
- An interrupt is enabled and the interrupted program is resumed.

Handling Multiple Devices:

When more than one device raises an interrupt request signal, additional information is needed to decide which device is to be considered first. The following methods are used to decide which device to select: Polling, Vectored Interrupts, and Interrupt Nesting.

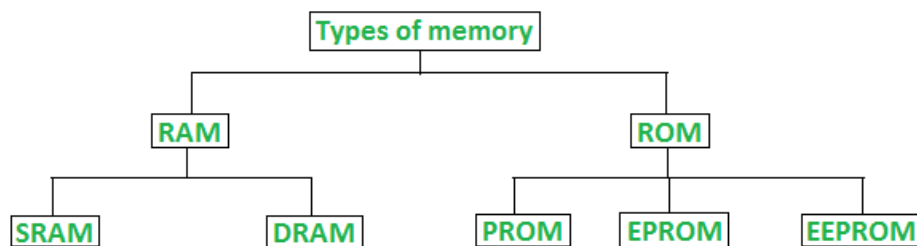
Polling: In polling, the first device that comes up with the IRQ is the device that is serviced first. The appropriate ISR is called to service the same. It is easy to implement but a lot of time is wasted in interrogating the IRQ bits of all the devices.

Vectored Interrupts: In vectored interrupts, a device requesting an interrupt identifies itself directly by sending a special code to the processor over the bus. This enables the processor to identify the device that generated the interrupt. The special code can be the starting address of the ISR or where the ISR is located in memory and is called the interrupt vector.

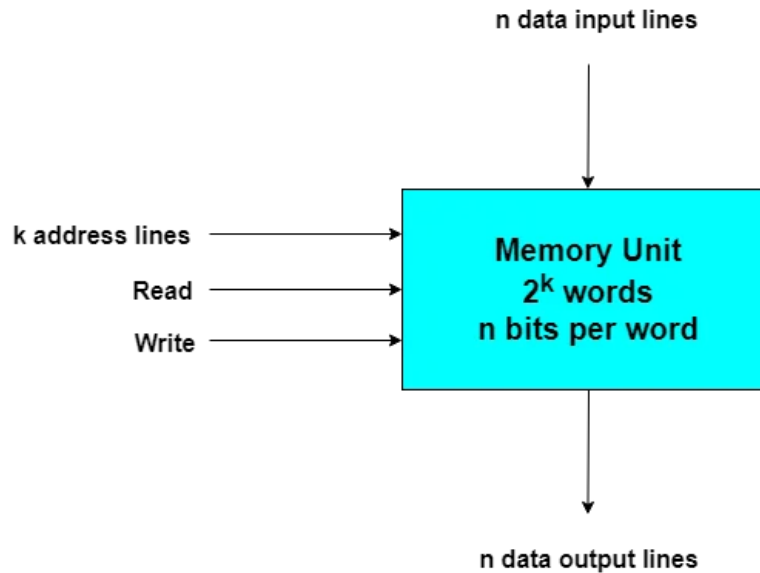
Interrupt Nesting: In this method, the I/O devices are organized in a priority structure. Therefore, an interrupt request from a higher priority device is recognized whereas a request from a lower priority device is not. The processor accepts interrupts only from devices/processes having priority.

Memory sub-system

Memory is the most essential element of a computing system because without it computers can't perform simple tasks. Both types of memory (RAM and ROM) are important for the computer, but they serve different purposes. RAM is used to store data that the computer is currently using, while ROM is used to store data that the computer needs to boot and operate. RAM is faster than ROM, as the data stored in it can be accessed and modified in any order, while data stored in ROM can only be read.



Classification of computer memory



Architecture of RAM

Random Access Memory (RAM)

- It is also called read-write *memory* or the *main memory* or the *primary memory*.
- The instructions and data that the CPU requires during the execution of a program are stored in this memory.
- It is a volatile memory as the data is lost when the power is turned off.

Advantages of Random Access Memory (RAM)

- **Speed:** RAM is much faster than other types of storage, such as a hard drive or solid-state drive, which means that the computer can access the data stored in RAM more quickly.
- **Flexibility:** RAM is volatile memory, which means that the data stored in it can be easily modified or deleted. This makes it ideal for storing data that the computer is currently using or processing.
- **Capacity:** The capacity of RAM can be easily upgraded, which allows the computer to store more data in memory and thus improve performance.
- **Power Management:** RAM consumes less power compared to hard drives, and solid-state drives, which makes it an ideal memory for portable devices.

Disadvantages of Random Access Memory (RAM)

- **Volatility:** RAM is volatile memory, which means that the data stored in it is lost when the power is turned off. This can be a problem for important data that needs to be preserved, such as unsaved work or files that have not been backed up.
- **Capacity:** The capacity of RAM is limited, and although it can be upgraded, it may still not be sufficient for certain applications or tasks that require a lot of memory.
- **Cost:** RAM can be relatively expensive compared to other types of memory, such as hard drives or solid-state drives, which can make upgrading the memory of a computer or device more costly.

Read-Only Memory (ROM)

- Stores crucial information essential to operate the system, like the program essential to boot the computer.
- It is non-volatile.
- Used in embedded systems or where the programming needs no change.
- Used in calculators and peripheral devices.
- ROM is further classified into four types- MROM, PROM, EPROM, and EEPROM.

Types of Read-Only Memory (ROM)

- 1. PROM (Programmable read-only memory):** It can be programmed by the user. Once programmed, the data and instructions in it cannot be changed.
- 2. EPROM (Erasable Programmable read-only memory):** It can be reprogrammed. To erase data from it, expose it to ultraviolet light. To reprogram it, erase all the previous data.
- 3. EEPROM (Electrically erasable programmable read-only memory):** The data can be erased by applying an electric field, with no need for an ultraviolet light. We can erase only portions of the chip.
- 4. MROM (Mask ROM):** Mask ROM is a kind of read-only memory that is masked off at the time of production. Like other types of ROM, mask ROM cannot enable the user to change the data stored in it. If it can, the process would be difficult or slow.

Advantages of Read Only Memory (ROM)

- **Non-volatility:** ROM is non-volatile memory, which means that the data stored in it is retained even when the power is turned off. This makes it ideal for storing data that does not need to be modified, such as the BIOS or firmware for other hardware devices.

- **Reliability:** Because the data stored in ROM is not easily modified, it is less prone to corruption or errors than other types of memory.
- **Power Management:** ROM consumes less power compared to other types of memory, which makes it an ideal memory for portable devices.

Disadvantages of Read Only Memory (ROM)

- **Limited Flexibility:** ROM is read-only memory, which means that the data stored in it cannot be modified. This can be a problem for applications or firmware that need to be updated or modified.
- **Limited Capacity:** The capacity of ROM is typically limited, and upgrading it can be difficult or expensive.
- **Cost:** ROM can be relatively expensive compared to other types of memory, such as hard drives or solid-state drives, which can make upgrading the memory of a computer or device more costly.

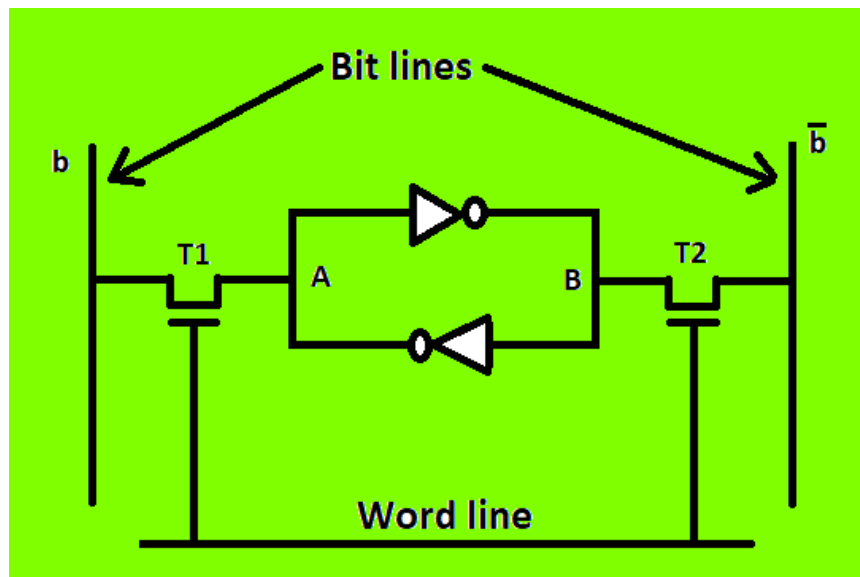
RAM	ROM
1. Temporary Storage.	1. Permanent storage.
2. Store data in MBs.	2. Store data in GBs.
3. Volatile.	3. Non-volatile.
4.Used in normal operations.	4. Used for startup process of computer.
5. Writing data is faster.	5. Writing data is slower.

Difference between RAM and ROM

SRAM stands for **Static Random Access Memory**. It is a form of a semiconductor. It is widely used in microprocessors, general computing applications and electronic devices. The SRAM is volatile in nature which means the data stored in it gets all wiped out once the power supply is cut. SRAM is comprised of flip flops. It consists of 4-6 transistors, once the flip flop stores the bit, it keeps it stored until the opposite bit is stored in it.

SRAM

Static memories (SRAM) are memories that consist of circuits capable of retaining their state as long as power is on. Thus this type of memory is called volatile memory. The below figure shows a cell diagram of SRAM. A latch is formed by two inverters connected as shown in the figure. Two transistors T1 and T2 are used for connecting the latch with two-bit lines. The purpose of these transistors is to act as switches that can be opened or closed under the control of the word line, which is controlled by the address decoder. When the word line is at 0-level, the transistors are turned off and the latch remains its information. SRAM does not require refresh time. For example, the cell is at state 1 if the logic value at point A is 1 and at point, B is 0. This state is retained as long as the word line is not activated.



For the **Read operation**, the word line is activated by the address input to the address decoder. The activated word line closes both the transistors (switches) T1 and T2. Then the bit values at points A and B can transmit to their respective bit lines. The sense/write circuit at the end of the bit lines sends the output to the processor.

For the **Write operation**, the address provided to the decoder activates the word line to close both switches. Then the bit value that is to be written into the cell is provided through the sense/write circuit and the signals in bit lines are then stored in the cell.

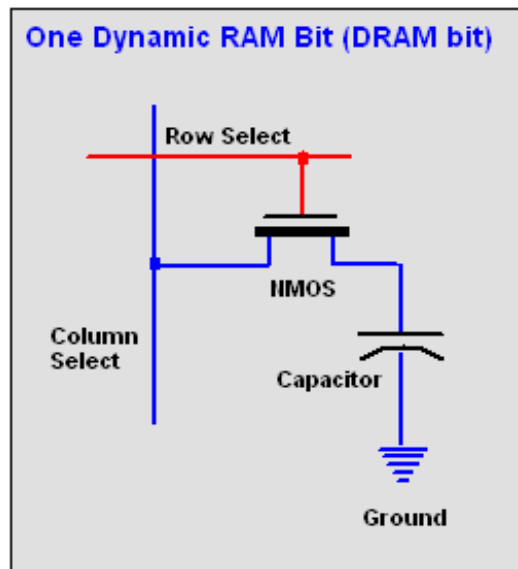
DRAM

DRAM stands for **Dynamic Random Access Memory**. It is a type of RAM that stores each bit of data (0 or 1) in a memory cell. These memory cells consist of two devices, a capacitor and a

transistor. Both are based on MOS (Metal Oxide Semiconductor) technology. The term dynamic is used because the DRAM changes its state from 0 to 1 over a period of time.

DRAM is relatively less expensive due to the use of one capacitor and one transistor in each cell. The information is stored in a DRAM cell in the form of a charge on a capacitor. This charge needs to be recharged periodically.

For storing information on the cell, the transistor is turned on and an appropriate voltage is applied to the bit line. This causes a known amount of charge to be stored in the capacitor. After the transistor is turned off, due to the property of the capacitor, it starts to discharge. Hence the information stored in the cell can be read correctly only if it is read before the charge on the capacitor drops below some threshold value.



SRAM	DRAM
SRAM has a lower access time, so it is faster compared to DRAM.	DRAM has a higher access time, so it is slower than SRAM.
SRAM is costlier than DRAM.	DRAM costs less compared to SRAM.
SRAM requires a constant power supply, which means this type of memory consumes more power.	DRAM offers reduced power consumption due to the fact that the information is stored in the capacitor.
Due to complex internal circuitry, less storage is available compared to the same physical size of a DRAM memory chip.	Due to the small internal circuitry in the one-bit memory cell of DRAM, a large storage capacity is available.
SRAM has low packaging capacity.	DRAM has a high packaging density.
SRAM does not require refresh time.	DRAM requires periodic refresh time.

Memory hierarchy

In computer system design, memory hierarchy is an enhancement to organize the memory such that it can minimize the access time. The memory hierarchy was developed based on a program behaviour known as locality of references.

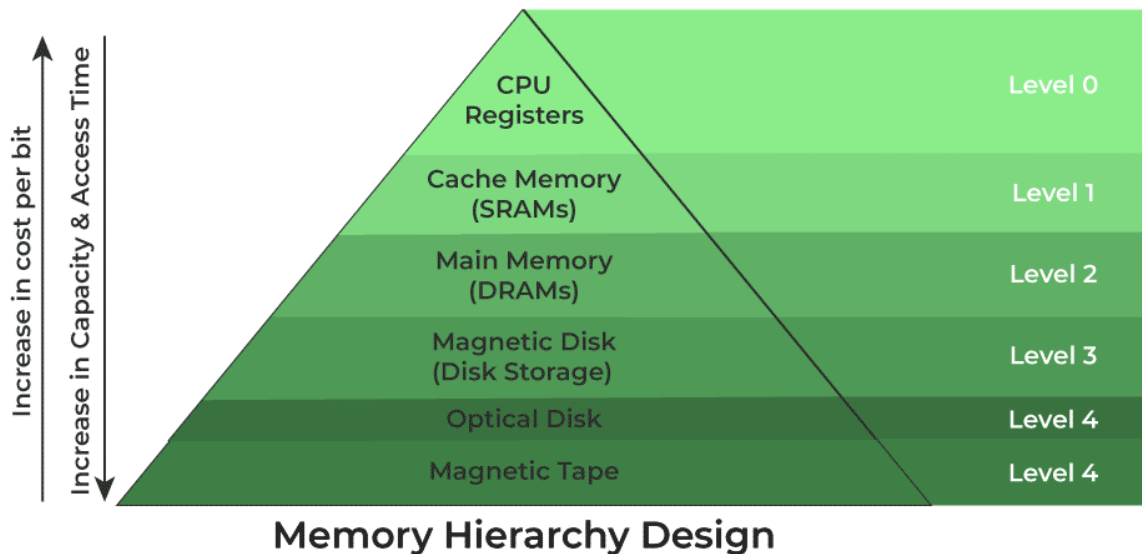
There are multiple levels present in the memory, with each one having a different size, cost, etc. Some types of memory like cache and main memory are faster compared to other types, but they are smaller and costlier whereas some types of memory are of bigger size, but are a little slower. Access to data is not similar in all types of memory, because some have faster and some have slower access.

Types of Memory Hierarchy

This Memory Hierarchy Design is divided into 2 main types:

- **Internal memory or primary memory:** Composed of main memory, cache memory and CPU registers. This is directly accessible to the processor.

- **External memory or secondary memory:** Composed of optical disk, magnetic disk and magnetic tape, i.e. peripheral storage devices which are accessible by the processor or via an I/O module.



Memory Hierarchy Design

1. Registers

Registers are small, high-speed memory units located in the CPU. They are used to store the most frequently used data and instructions. Registers have the fastest access time and the smallest storage capacity, typically ranging from 16 to 64 bits.

2. Cache Memory

Cache memory is a small, fast memory unit located close to the CPU. It stores frequently used data and instructions that have been recently accessed from the main memory. Cache memory is designed to minimize the time it takes to access data by providing the CPU with quick access to frequently used data.

3. Main Memory

Main memory, also known as RAM (Random Access Memory), is the primary memory of a computer system. It has a larger storage capacity than cache memory, but it is slower. Main memory is used to store data and instructions that are currently in use by the CPU.

Types of Main Memory

- **Static RAM:** Static RAM stores the binary information in flip flops and information remains valid until power is supplied. It has a faster access time and is used in implementing cache memory.
- **Dynamic RAM:** It stores the binary information as a charge on the capacitor. It requires refreshing circuitry to maintain the charge on the capacitors after a few milliseconds. It contains more memory cells per unit area as compared to SRAM.

4. Secondary Storage

Secondary storage, such as hard disk drives (HDD) and solid-state drives (SSD), is a non-volatile memory unit that has a larger storage capacity than main memory. It is used to store data and instructions that are not currently in use by the CPU. Secondary storage has the slowest access time and is typically the least expensive type of memory in the memory hierarchy.

5. Magnetic Disk

Magnetic disks are simply circular plates that are fabricated with either a metal or a plastic or a magnetized material. The Magnetic disks work at a high speed inside the computer and these are frequently used.

6. Magnetic Tape

Magnetic Tape is simply a magnetic recording device that is covered with a plastic film. It is generally used for the backup of data. In the case of a magnetic tape, the access time for a computer is a little slower and therefore, it requires some amount of time for accessing the strip.

Characteristics of Memory Hierarchy

- **Capacity:** It is the global volume of information the memory can store. As we move from top to bottom in the Hierarchy, the capacity increases.
- **Access Time:** It is the time interval between the read/write request and the availability of the data. As we move from top to bottom in the Hierarchy, the access time increases.
- **Performance:** Earlier when the computer system was designed without a Memory Hierarchy design, the speed gap increased between the CPU registers and Main Memory due to a large difference in access time. This results in lower performance of the system and thus, enhancement was required. This enhancement was made in the form of Memory Hierarchy Design because of which the performance of the

system increases. One of the most significant ways to increase system performance is minimizing how far down the memory hierarchy one has to go to manipulate data.

- **Cost Per Bit:** As we move from bottom to top in the Hierarchy, the cost per bit increases i.e. Internal Memory is costlier than External Memory.

Advantages of Memory Hierarchy

- It helps in removing some destruction, and managing the memory in a better way.
- It helps in spreading the data all over the computer system.
- It saves the consumer's price and time.

Disadvantages of Memory Hierarchy

The memory hierarchy design provides a solution to optimize the overall performance of a system while balancing the cost. However, it also has some disadvantages:

- **Complexity:** It adds complexity to the system as there are multiple layers of memory, and the system needs to manage them efficiently.
- **Access time:** Although the memory hierarchy design aims to reduce the access time to the data, it still adds some latency in the system. The data needs to move from one layer to another, which increases the access time.
- **Cost:** While memory hierarchy design aims to balance the overall cost, it still adds to the cost of the system. The higher the number of layers, the more costly the system becomes.
- **Limited capacity:** Each layer of the memory hierarchy has limited capacity, which means that the system cannot store an unlimited amount of data. This limitation can sometimes lead to performance issues.
- **Management overhead:** The system needs to manage the data movement between the different layers of memory, which adds overhead to the management process.

Despite these disadvantages, memory hierarchy design is still widely used as it provides an efficient way to manage memory and optimize system performance.